# Training-free MDS form filling from discharge summaries

*MDS 2.0 Sections I (diagnoses), O (medications), P (special treatments) from MIMIC-IV notes and hospital records. Results so far, challenges, and the design we would use.*

## 1. The ideas in brief

- **Admission knowledge graph.** A model extracts condition, drug and treatment mentions from the note, each with who it is about, whether it is affirmed or negated, and when it applies; a verifier checks them. Dated hospital records (diagnosis codes, prescriptions, administrations, ICU events) join the same graph, the model maps every concept to form items from the item wording alone, and the evidence path that fills an item is the explanation.
- **No predefined rules, no training.** One prompt per component, reused unchanged across datasets, a second language (English to Italian) and a second form (MDS 2.0 to 3.0).
- **Direct whole-form prompt as the baseline decider,** plus agreement and tie-break checks where methods disagree (an NLI model as abstention verifier, GLiNER as an NER consistency check).
- **Label-free robustness tests** (mask, negate, paraphrase, reorder and distractor edits of the cited evidence), first checked against labelled accuracy on NegEx before use on MIMIC, where no gold exists.

## 2. Results achieved

| Test (public data) | Our training-free result | Comparison |
|---|---|---|
| **Negation + timing, NegEx (2,376 mentions)** | Status record: negation F1 0.976, timing macro-F1 0.820 | ConText rules 0.929 / 0.798 |
| **Medication extraction (glaucoma notes)** | Model NER + verifier F1 0.924 (400 lines) | Trained CRF 0.938; GLiNER 0.528 |
| **Diagnosis coding ranking (CodiEsp-D, 250 cases)** | Training-free MAP 0.585, about 2nd of 22 teams | Trained graph MAP 0.619 (1st) |
| **Form filling, 134-item CRF, English test (200 notes)** | Direct whole-form prompt 0.663; graph + audit 0.607 (macro-F1) | Best team 0.683; baseline 0.470 |
| **Same, Italian test, unchanged code** | Direct 0.669; graph + audit 0.622 | Best team 0.672 |
| **Same, open model (Qwen3.8-27B)** | Direct 0.561; agreement with form-guided NER 0.620 (+0.059, CI 0.036 to 0.083) | Open model over-fills: 19 values per note vs about 5 in the gold |
| **MIMIC-IV admissions, no gold labels (label-free tests)** | Ignores family-history and ruled-out distractors 100%; stable under paraphrase 0.94, reordering 0.97, negation 0.90; MDS 3.0 form gives the same answers 97% | Evidence completeness when masked: graph 0.84 vs reader 0.60; $0.008 to 0.011 and 33 to 52 s per admission |

*Placements are indicative: different models on the same test sets. Ablation on the CRF: model-decided item membership beats encoder membership (+0.055 with gpt-6-luna, +0.19 with Qwen); the consistency audit adds +0.03 with gpt-6-luna and nothing with Qwen; the verifier and the status record show no clear effect on filled items.*

## 3. Challenges

- **No gold MDS labels.** MIMIC accuracy is unknown; we have agreement and robustness only. A labelling app for nurses is built; their labels become the reference.
- **The graph over-fills and loses to the direct prompt on short forms** (0.607 vs 0.663): it decides each item without seeing the whole form. Encoder, NLI and GLiNER checks hurt as hard gates or equal-weight scores; they help only as tie-breakers.
- **Look-back and records.** Windows are counted from hospital discharge, not from a nursing-home assessment; notes rarely state medication days, so Section O comes from records (graph 2.9 drug classes per admission vs 0.75 from the note alone).
- **Model dependence and small samples.** Combining methods helped an over-filling model (+0.03 to +0.06) but not a well-calibrated one (+0.004). Several ideas were tried on 20 to 80 notes; the English CRF test set was looked at four times.

## 4. Final design (polished)

**Decide with the direct whole-form prompt;** the graph supplies cited evidence paths and the records that notes lack; **if the model over-fills** (values set per note well above the dev expectation), keep only values a second method agrees with or the NLI/NER tie-break supports; **everything still in disagreement goes to a nurse review queue** with highlighted evidence. On the CRF, values all methods agree on are 82% of fills at 0.63 precision, while two-method-only agreement drops to 0.17, so agreement is the triage signal. Cost: direct about $0.0005 and 5 s per CRF note; on MIMIC direct about $0.0013 and 14 s per admission, the graph about $0.011 and 52 s. Next: nurse labels on 100 admissions to measure real accuracy of direct, graph and their agreement.